From Words to Coalitions: Public vs Private Communication in Bargaining Game with LLM Text Analysis.

Luca Delle Foglie, [1,2], Stefano Papa, [1,2], Donato Pierno, and [1,2], and James Tremewan [3]

[1] Department of Economics and Finance, Tor Vergata University of Rome, Rome, Italy
[2] Centre for Investigation and Modelling of Experimental Observations (CIMEO), Sapienza University of Rome, Rome, Italy
[3] HSE University, Moscow, Russia

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1 Introduction

We study how communication protocols shape coalition formation in a bargaining game. Does limiting discussion to public communication lead to an all-inclusive alliance? Does allowing private communication foster the smallest feasible alliance? Is private communication necessary for majority decision-making to displace prosocial outcomes? Do messages differ between different communication protocols? What kinds of messages are effective in persuading people to form coalitions? We address these questions with a laboratory experiment.

Our experiment provides insights into the frequent claims about public and private communication in collective decisions, and this phenomenon is observed under the effect of heterogeneous "disagreement payoffs", and for the first time in the bargaining literature, we elicit individual's ambiguity-neutral beliefs for natural events (i.e., the opponent's choice).[1] In this paper, we conduct a multilateral bargaining experiment with free-form one-shot message. We manipulate two characteristics of our bargaining game: first, communication (*Public vs Private*), and second disagreement payoffs (*Total and No deadweight loss*). In our experimental game, participants are divided in groups of three (hereafter triad), and required to make a decision, among three predefined options proposed by the experimenter, about the allocation of an exogenous surplus. This setting enables participants to choose among a two-way split with only one participant (*unequal* allocation) and even three-way split (*equal* allocation). In this environment the allocation that receive at least two votes (majority rule), is selected for real payment. In our multilateral game, we permit agreements to include the smallest feasible alliance (i.e., including only a simple majority) or the all-inclusive alliance (i.e., including all the members of the triad). According with the terminology used in the literature on multilateral bargaining (e.g. Eraslan and Evdokimov, 2019; Baranski et al., 2024), we refer to these type of alliance as minimum winning coalitions (MWCs) and grand coalitions (GCs), respectively. Furthermore, we categorize and analyze the content of the communication occurred in our experiment employing innovative techniques of text analysis using Large Language Models (LLMs). Gentzkow et al. (2019), in their contribution provide an extensive introduction to the use of text in economic research, and the statistical method used up to that time. In this experimental paper, we decide to move a step ahead in the direction pioneered by Celebi and Penczynski (2024), which study the use of the Generative Pre-trained Transformer (GPT) series (Brown et al., 2020), for text classification in experimental economics, as a first evidence in the literature. Multiple

[1] Miller et al. (2018)is the closest experimental paper to our study, which study legislative bargaining game with heterogeneous disagreement payoff under majority and unanimity rule. Baron et al. (2017) Study the effect of communication protocol either private and public.

studies in different economic fields (i.e., Macroeconomics: Alonso-Robisco and Carbó, 2023; Smales, 2023; Hansen and Kazinnik, 2024; Fanta and Horvath, 2024. Finance: Lopez-Lira and Tang, 2023; Jha et al., 2024) have previously suggested that GPT models outperform existing text classification techniques such as Bidirectional Encoder Representations from Transformers (BERT, Devlin et al., 2019).

2 Related literature

At first glance, communication seems to play no role in bargaining games with complete information. In these games, two or more players must divide a resource under rules that specify who proposes, how proposals reach a vote, and how many votes are required for passage. Recent findings on two-person bargaining suggest the existence of the "power of asking": when players can communicate, allocations tend to be more egalitarian.[2] On the other hand, Agranov and Tergiman (2014) and Baranski and Kagel (2015)–comparing bargaining outcomes in the Baron-Ferejohn (hereafter BF) bargaining-game (Baron and Ferejohn, 1989)–suggest that as the bargaining group grows, the effect flips. Divergent interest among the group's components leads to not credible communication driven by conflict of interest (Schelling, 1957). In multilateral bargaining with majority rule, communication pushes outcomes away from egalitarian splits. Another remarkable result is that most approved allocations are MWCs: just enough members receive positive shares to secure a majority, while the rest receive nothing.[3] The Baron–Ferejohn (BF) model has attracted sustained attention for theory, see Eraslan et al. (2022); for a meta-analysis of structured multilateral bargaining experiments, see Baranski and Morton (2022); for the experimental literature, see Agranov (2022). The experimental evidence closest to our study is by Baron et al. (2017), who run a dynamic divide-the-dollar experiment to study how free-form communication—public or private—shapes bargaining outcomes and the formation and stability of coalitions. The game features an endogenous status quo; in each round, one member is randomly selected to propose an allocation of a divisible resource. A proposal passes with at least two affirmative votes (simple majority); otherwise, the status quo prevails. Before proposals and voting, participants may exchange unrestricted cheap talk. The authors compare three committee regimes: no communication; public communication (all messages observable); and private communication (members can message one another privately). They suggest, first, how three-person groups allocate resources across these regimes and, second, whether communi-

[2]Evidence comes from different studies employing the simplest dictator game (Mohlin and Johannesson, 2008; Andreoni and Rao, 2011; Ellingsen and Johannesson, 2008).

[3]For a detailed discussion of these topics, see the chapter titled "Communication in Games" in the *Handbook of Experimental Methodology*.

cation changes the emergence and duration of coalitions. Finally, they explore the content of the conversations to shed light on the mechanism behind the two above-mentioned outcomes (resource allocation and coalition formation). Baron et al. (2017) finds that private communication increases majoritarian allocations—especially equal two-member splits—and reduces universal allocations.[4] Conversely, public communication makes majoritarian allocations less likely and universal allocations more likely. In their data, equal (universal) splits are far less common with private than with public communication (45% vs. 95%), and lie in between with no communication (71%). The largest difference is between private and public (50 percentage points). By contrast, majoritarian (unequal) allocations are most frequent under private communication and rarest under public communication (4%).

3 Experimental Design

Our between subject experiment, consist of three experimental parts. The first part of the experiment consists of playing a bargaining game, the second part is designed to elicit individuals' ambiguity aversion and ambiguity generated insensitivity by means of the matching probability methodology developed by Baillon et al. (2018), and finally, the third part is designed to elicit individual measures of participants' social preferences through a dictator game. The first and third parts consist of dividing an exogenous surplus of €12 among a group of three people, where each participant could earn between a minimum of €0 and a maximum of €6. We use a stranger matching protocol to ensure that each participant plays each of the two games with two new participants, to prevent and minimise any order effect and influence between parts one and three.[5] At the beginning of the experiment, each subject is assigned a fictitious identity of letter and number (i.e., Participant B3 or Participant F2); each of the three parts is presented and explained to our participant one next to the other, after finishing the previous one. At the end of the experiment, each participant is informed about the earned payoff, which is composed of a fixed participation fee of €2 plus the payoff earned from part two, and either the payoff from part one or part three, which is randomly selected at the group level. Instructions together with the relevant control questions for each experimental part are presented on the screen to our participants. Part one

[4]They define a *majoritarian* allocation as one that assigns a significant share of tokens to exactly two committee members—i.e., an unequal resource allocation. However, they define a *universal* allocation as one that equally splits the assigned tokens among the three participants—i.e., an equal resource allocation.

[5]In more detail, if participants remained in the same groups for both the bargaining and dictator games, their decisions in the dictator game could be influenced by a desire to reward or punish others based on anticipated outcomes of the bargaining game, inferred from prior conversations and received signals. This interdependence would compromise the ability to obtain an independent measure of social preferences in the dictator game.

is a multilateral bargaining game played with free-form one-shot message; each subject has to divide the surplus between themselves and two other participants. The division of the surplus occurs through three options, equal for each subject. An option that receives at least two votes (MWCs) is implemented for the group. The options are a two-way split with only one participant (i.e. (6,6,0) and (6,0,6), *unequal* allocation) and even three-way splits (i.e. (4,4,4), *equal* allocation). Part two consists of a series of 12 questions which are relatives to the others participant choice after part one. In each question, subjects are asked to choose between two options. Part three is a three-player dictator game where each subject has to choose how to divide the given surplus between themselves and two other participants; each participant has three allocation alternatives proposed by the experimenter, equal for each subject, either in parts one and three. In order to enable subjects to interact flexibly and quickly, the experiment is programmed with a computer interface using the zTree software (Fischbacher, 2007).

3.1 Part One: Bargaining Game

This part is the main focus of the experiment. Through an unstructured one-shot bargaining game, each participant is randomly assigned to a triad (each participant belongs only in one triad), and has to divide the surplus of €12.[6] Participants in the triads are labelled with a fictitious identity to ensure anonymity (i. e., Participant A1, B1, C1). In our design, each participant has three predefined options to distribute the surplus: a two-way split, i.e. (6,6,0) and (6,0,6); and additionally a three-way split, i.e. (4,4,4). Nonetheless, a majority rule (i.e. two or more participants agree on the same distribution option) is used to determine the majority's choice to implement for real payment. However, if no majority is reached (i.e., each participant selects a different option), the group fail to coordinate, and all three participants earn €0 (baseline disagreement payoffs *Total Deadweight Loss*). We propose two alternative mechanisms to determine the disagreement payoffs (see subsections 3.4). The playing screen is divided into two independent bilateral chat boxes, simultaneously active for a total of three minutes. During this time, they have to communicate in simultaneous, separate two-way conversations. For example (see Figure 1), Participant B1 will have two chat windows open on their screen: one on the left to chat with Participant A1, and one on the right to chat with Participant C1. At the end of the communication period, before proceeding with the redistribution decision, each participant is required to send a pre-written message (hereafter signal) to each of the other two participants. For instance, Participant A1, after

[6] Crucial, our experiment builds on recent studies of unstructured bargaining games such as Bolton et al. (2003); Bolton and Brosig-Koch (2012); Camerer et al. (2019) Tremewan and Vanberg (2020); Baranski et al. (2024); Kamm and Siegenthaler (2024).

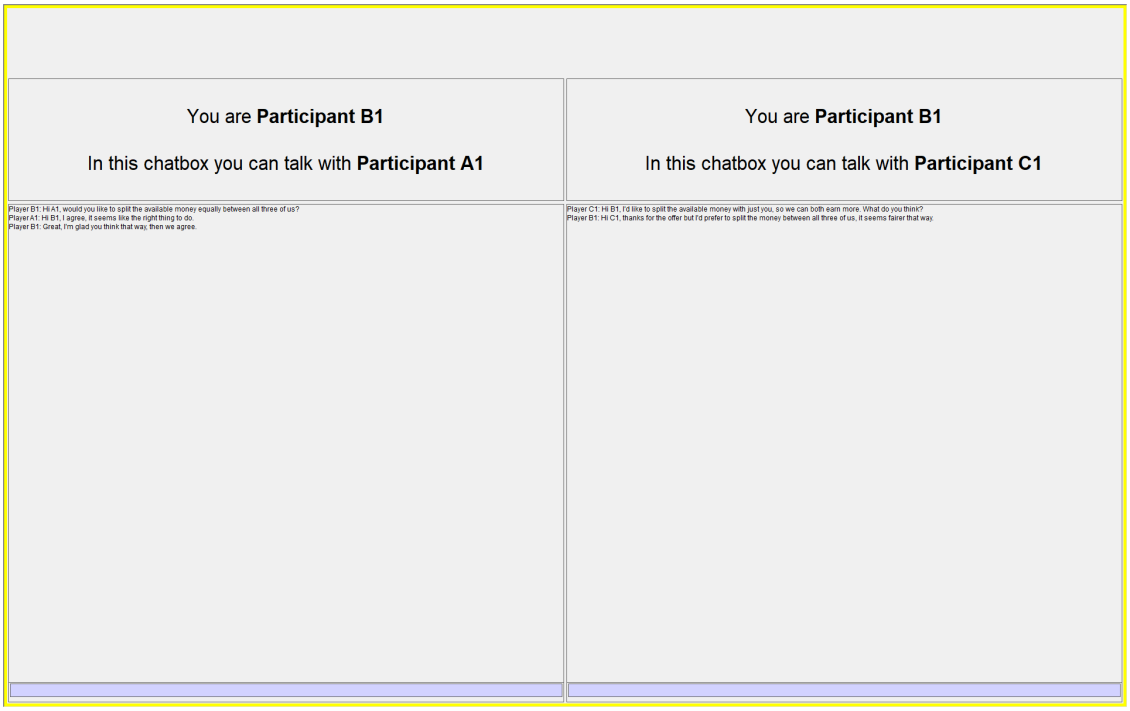

Fig. 1: Chat Screen

the communication ends, may choose one of the following pre-written messages to Participant B1 (C1):

- I wish to split the payoff equally with Participant B1 only.
- I wish to split the payoff equally with Participant C1 only.
- I wish to split the payoff equally with both of the other participants.
- I do not wish to communicate my intentions.

Finally, after choosing one of the signals, each participant makes their redistribution choice. Before making their choice, participants in the baseline group (*Public communication*), exactly after the free-form one-shot message, are able to read for 90 seconds all the messages that occurred within the triad (i.e. Participant A1 is able to read the conversation that occurred between Participant B1 and C1, and vice versa). Participants in the treated group (*Private communication*) make their decision in the complete ignorance of the communication that occurred between the other two participants (i.e. Participant A1 is not allowed to read the conversation that occurred between Participant B1 and C1, and vice versa).

3.2 Part Two: Matching Probability

This part of the experiment consists of eliciting individual ambiguity aversion, ambiguity-generated insensitivity and ambiguity-neutral beliefs for natural events, through the use of the matching probability procedure presented by Baillon et al. (2018). Each participant is presented with 12 questions, 6 for each participant in the triad. Each question measures the participant's preference between betting on given probabilities and betting on the choices of other participants in part one (see Figure 2). The latter are referred to either single, and composite events (see Table 1, and Figure 3). For example, suppose you are participant A1, part of the triad with participants B1 and C1 in Part One; each question is presented with two options:

- **Option 1:** You win €5 if Participant B1 chose "Share only with Participant C1" or "Share with both Participant A1 and Participant C1" (and nothing otherwise).
- **Option 2:** You win €5 with p% probability (and nothing otherwise).

In all questions, each subject is offered the chance to win €5, the probabilities range of the above options span from 0 to 100. These questions are presented in the form of multiple price lists (MPLs) with ascending p values, and the participant is provided with an on-screen reminder of the signals previously sent and received.

3.3 Part Three: Dictator Game

In the three-person dictator game, each subject is randomly assigned to a new triad (i.e., Participant A1 is grouped with Participant D1 and Participant H1), and is free to divide a surplus of €12 between themselves and the other subjects in the triad (i. e., each participant acts as a dictator). Each participant has three predefined options for how to distribute the surplus: two-way splits with only one participant (*unequal* allocation), or three-way splits between all three (*equal* allocation). Equal allocations redistribute €4 for each triad's member. Unequal allocations redistribute €6 to the player in the dictator role and only one of the two other participants. Each triad member's decision was equally likely to be selected for the purpose of calculating payments. If the final payout is determined by the dictator game, one of the three participants' choices within the triad will be randomly selected and implemented, with each having an equal probability of being selected to be the dictator. Communication was not present in this part, and participants did not learn about each other's choices from Part One until the end of the experiment.

OPTION 1	1	2	OPTION 2
You win € 5 if Participant A1 chose **Share with both Participant B1 and Participant C1** (and nothing otherwise)			**You win € 5** with the following probability (and nothing otherwise)
We remind you of the following information regarding what you and Participant A1 have declared to each other: You said: I want to share only with Participant B1. Participant A1 said: I want to share with both Participant B1 and Participant C1	■	□	0%
	■	□	1%
	■	□	2%
	■	□	5%
	■	□	10%
	■	□	15%
	■	□	20%
	□	■	25%
	□	■	30%
	□	■	35%
	□	■	40%
	□	■	45%
	□	■	50%
	□	■	55%
	□	■	60%
	□	■	65%
	□	■	70%
	□	■	75%
	□	■	85%
	□	■	100%
			OK

Fig. 2: Example of an MPL regarding a single event.

OPTION 1	1	2	OPTION 2
You win € 5 if Participant B1 chose **Share only with Participant C1 or Share with both Participant A1 and Participant C1** (and nothing otherwise)			**You win € 5** with the following probability (and nothing otherwise)
We remind you of the following information regarding what you and Participant B1 have declared to each other: You said: I want to share with both Participant B1 and Participant C1 Participant B1 said: I want to share with both Participant A1 and Participant C1	■	□	0%
	■	□	20%
	■	□	35%
	■	□	40%
	■	□	45%
	■	□	50%
	■	□	55%
	■	□	60%
	■	□	65%
	■	□	70%
	■	□	75%
	■	□	80%
	□	■	85%
	□	■	90%
	□	■	93%
	□	■	95%
	□	■	97%
	□	■	98%
	□	■	99%
	□	■	100%
			OK

Fig. 3: Example of an MPL regarding a composite event.

3.4 Treatments and Session Details

Our main treatment is the way in which communication is allowed: *public* (baseline communication), and *private* (treatment communication). In order to compare the effect of showing private messages on coalition formation, our communication environment is structured as follows: each participant has three minutes to send a one-shot message to the other participants in the triad. The communication occurs through two separate, simultaneous chat windows (hereafter chat-box) displayed on the screen (positioned one on the left and one on the right, see Figure 1). Each chat-box connects only two participants (i.e., Participant A1 has one chat-box for each of the other triad participants B1 and C1). The main characteristic of our treatment (*private* communication) lies in the fact that each participant can only view the two-person message they are part of, and the content of the message between the other triad participants remains completely hidden. What differentiates our treatment from the baseline group is that the latter, shows the hidden conversation in which each participant does not take part. In other words, participants in the baseline group can also read conversations in which they are not directly involved (i.e., participant A1 can read the conversation that occurred between B1 and C1). The same treatment characteristic is applied to the signal each participant sends to the other triad members. Finally, in order to measure the effect of different disagreement payoffs on coalition formation under private communication protocol, we observe subjects' behaviour under two different mechanisms: *Total Deadweight Loss* (TDL, baseline disagreement payoff), and *No Deadweight Loss* (NDL, treatment disagreement payoff). Our baseline disagreement payoff assigns to each subject €0 if no majority is reached (see Table 2). In this mechanism, a subject maximises their payoff by successfully reaching an agreement with another participant, while excluding the third party.[7] However, the difference in the treatment disagreement payoff lies in the fact that when no majority is reached, each subject could earns a different payoff (heterogeneous payoff among the triad members) based on the combination of his or her choice and that of the other participants (see Table 3). If no majority is reached, all three participants could earn €2 or €4 or €6 or €8. In this context, a subject maximizes their payoff by successfully reaching an agreement on an equal split with both participants, while choosing to divide equally among the three.[8]

[7]Table 2 reports the game in normal form under total deadweight loss, highlighting the Nash equilibria.
[8]Table 3 reports the game in normal form under no deadweight loss, highlighting the Nash equilibria.

	TDL	NDL
Public	✕	
Private	✕	✕

4 Hypotheses

Our hypotheses are motivated by existing evidence from the BF game. The differences between our less structured bargaining process and the BF model, guide us in state our hypotheses based on the previous findings from existing work.

Woon et al. (2024) suggest that public communication may elicit or clarify social norms about appropriate behaviour in a given context, similar to communication inducing cooperative behaviour in social dilemmas (Bicchieri and Lev-On, 2007). In our simultaneous one-shot bargaining game with two unequal allocation and one equal, an alternative is implemented only if it receives at least two votes, otherwise the disagreement payoff is the earning of all group participant.

To analyze the NDL case, we draw inspiration from level-k theory (Stahl II and Wilson, 1994; Nagel, 1995; Stahl and Wilson, 1995). Considering a subject devoid of social preferences,[9] we expect that, if she is a level-0 player, she will attempt to implement the strategy that maximizes her own payoff without accounting for the possibility that others may anticipate her actions.[10] Thus, a selfish level-0 player A will adopt a strategy profile $S_A : \{c_A : \textit{share with both } B \textit{ and } C; s_{A,B} : \textit{share only with } B; s_{A,C} : \textit{share only with } C\}$, where c_A denotes A's choice, $s_{A,B}$ the signal sent to player B, and $s_{A,C}$ the signal sent to player C. This strategy reflects an attempt to deceive others, without recognizing that such behavior is easily anticipated. If all members of a group are level-0 players lacking social preferences, a GC with payoff distribution $\{\pi_A; \pi_B; \pi_C\} = \{4; 4; 4\}$ will emerge due to the complete failure

[9] A subject characterized by social preferences, regardless of her degree of strategic sophistication, would propose to the other players and opt to distribute the available amount equally among all group members.

[10] In the literature on level-k reasoning, particularly in the context of the beauty contest game, a level-0 player is typically defined as one who makes a purely random choice. In our setting, however, we assume that all participants are capable of identifying the combination of choices within the group that would maximize their payoff—given that they engage in the chat phase and subsequent decision-making only after receiving thorough instructions and answering control questions. Accordingly, we define a level-0 player as one who pursues the strategy that yields the highest payoff, disregarding the predictability of this behavior and making no inferences about the strategic reasoning of others as .

of deceptive strategies. A selfish level-1 participant, recognizing that the maximum payoff of €8 is unattainable under these conditions, will anticipate the default outcome of $\{4; 4; 4\}$ in the absence of effective communication. Consequently, she will attempt to persuade another participant to form a MWC, aiming for a payoff of €6. For example, player A, assuming level-1 reasoning and believing the others to be level-0, understands that forming a MWC incurs no cost. If the others are indeed level-0, her efforts will not alter the game's outcome. However, if she succeeds in persuading another selfish player to elevate their reasoning from level-0 to level-1 during the chat phase, they can jointly achieve a payoff of €6. At this stage, it becomes intuitive that a selfish level-2 player A perceives an opportunity to reach the maximum payoff of €8. She anticipates that level-1 players will actively seek to form MWCs, and thus she can implement the strategy $S_A : \{c_A : \textit{share with both } B \textit{ and } C; s_{A,B} : \textit{share only with } B; s_{A,C} : \textit{share only with } C\}$, attempting to convince both players B and C that she is the optimal partner for achieving a payoff of €6. However, as with a group composed primarily of level-0 players, the presence of two or more level-2 players will lead to a convergence toward the distribution $\{4; 4; 4\}$. Finally, a level-3 player A will behave similarly to a level-1 player. Based on the reasoning outlined for levels 0 through 3—and without extending the analysis further, given that the literature indicates most experimental subjects fall within this range (e.g., Crawford et al., 2013)—we now consider how these dynamics interact with the presence or absence of deadweight loss resulting from disagreement. Assuming that the distribution of level-k reasoning is consistent across the two treatments, TDL and NDL, we expect that in the NDL treatment, level-1 and level-3 subjects will be more inclined to attempt forming a MWC due to the reduced cost associated with disagreement payoffs. Level-2 players will benefit from this situation strategically, while level-0 players will benefit more passively. Therefore, assuming equal proportions of level-0/2 and level-1/3 players in both treatments, we anticipate that under private communication, the NDL treatment will exhibit a higher percentage of groups failing to coordinate compared to the TDL treatment (due to the fact that not reaching an agreement in TDL treatment is costly, assigning a payoff equal 0 rather than a positive amount greater than the latter).

In the case of the NDL, we see that the number of Nash equilibria is significantly smaller (Table 3). It is expected that, the subjects will tend to converge to the equilibrium in which everyone chooses to divide equally among all three players. However, they are expected to reach this equilibrium differently. Imagine that player A implements a strategy profile $S_A : \{c_A; s_{A,B}; s_{A,C}\}$ where c_A is the A's choice, $s_{A,B}$ is the signal sent to player B, and $s_{A,C}$ is the signal sent to player C. In the presence of a private communication, player A is expected to implement $S_A : \{ABC; AB; AC\}$; players B and C are expected to behave similarly, thus converging to the payoff distribution $\{4, 4, 4\}$. Given this, and our two treatment relative to

the disagreement payoff mechanisms, we state the following:

Hypothesis 1a: Under TDL , the proportion of groups failing to coordinate (i.e., each participant selects a different income distribution) is greater under private communication compared to public communication.

Hypothesis 1b: Under private communication, the proportion of groups failing to coordinate (i.e., each participant selects a different income distribution) is greater in the NDL treatment compared to the TDL treatment.

The formation of a minimum winning coalition against a grand coalition, is certainly influenced by the type of communication allowed during the bargaining game. Baron et al. (2017), suggest experimental evidence that private communication increases coordination on both unequal coalitions and allocations, while decreases coordination on both equal coalitions and allocations, whereas public communication has the opposite effect. This evidence comes from the classical BF model rather than our less structured bargaining process. We state the hypotheses 2a and 2b as follow:

Hypothesis 2a: Under TDL, the proportion of groups choosing the triadic equal split ($\{\pi_A; \pi_B; \pi_C\} = \{4; 4; 4\}$, equal allocation) is greater under public communication compared to private communication.

Hypothesis 2b: Under private communication, the proportion of groups choosing the triadic equal split ($\{\pi_A; \pi_B; \pi_C\} = \{4; 4; 4\}$, equal allocation) is greater under NDL treatment compared to the TDL treatment.

During the free-form one-shot message, both private and public, the screen in our experiment is divided into two independent bilateral chat boxes, simultaneously active for three minutes. Being aware that most of languages require reading and writing from the left to right, and according to experimental evidence on the effect of position on screen on subjects' choice order (Cason et al., 2012),[11] subjects are more likely to communicate and reach agreements with the partner on the left side of the screen. Specifically our conjecture is the following:

Hypothesis 3: Most first movers (the first person to send a message within a group) will send the first message in the chat on the left side of the screen.

[11] In this experiment, participants simultaneously engaged in two games displayed on opposite sides of the screen. The authors observed that, irrespective of which game was presented on the left/right side, over 90% of participants submitted their choice for the game presented on the left side first, followed by the game presented on the right side.

The main characteristics of our experiment, lies in two type of communication: Public and Private. We hypothesize that making conversations public—thereby eliminating the possibility of deception-based strategies and enhancing the credibility of others' statements—will result in more polarized beliefs regarding others' choices. In this context, we measure the distribution of beliefs about others' choices at the individual level based on the matching probabilities elicited in Part 2. Specifically, for each participant, we compute the probabilities assigned to the partner's possible choices in Part 1—$p(Option_1)$, $p(Option_2)$, $p(Option_3)$—which are non-negative and sum to one.[12] To quantify the asymmetry in an individual's belief distribution, we plan to rely on the Euclidean distance between the belief vector $\big(p(Option_1), p(Option_2), p(Option_3)\big)$ and the uniform distribution vector $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. This metric captures the deviation from a neutral prior, where all options are considered equally likely.

Hypothesis 4: Public communication leads to greater asymmetry in beliefs about others' choices in Part 1 compared to private communication.

According to Erazo and Kpegli (2024), greater belief asymmetry is associated with reduced ambiguity aversion and increased ambiguity generated insensitivity. Based on this theoretical and experimental evidence, we state our hypotheses:

Hypothesis 5: The index of ambiguity aversion is greater under private communication compared to public communication.

Hypothesis 6: The index of ambiguity generated insensitivity is greater under public communication compared to private communication.

[12] These beliefs, representing the perceived likelihood of mutually exclusive events, are calculated using the method introduced by Li et al. (2019).

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Appendices

Appendix A Tables

A.1 Matching Probability events

The following events listed in Table 1, are the ones used during part two.

A.2 Total Deadweight Loss

If no majority is reached, all three participants earn €0. The payoffs for all the participants are reported in Table 2.

A.3 No Deadweigth Loss

If no majority is reached, all three participants could earn €2 or €4 or €6 or €8. The payoffs for all the participants are reported in Table 3.

Participant	Event	Description
A	E_1^B	Participant B divide equally with Participant A only
	E_1^C	Participant C divide equally with Participant A only
	E_2^B	Participant B divide equally with Participant C only
	E_2^C	Participant C divide equally with Participant B only
	E_3^B	Participant B divide equally among all three participants
	E_3^C	Participant C divide equally among all three participants
	E_{12}^B	Participant B divide equally with Participant A only *or* Participant B divide equally with Participant C only
	E_{12}^C	Participant C divide equally with Participant A only *or* Participant C divide equally with Participant B only
	E_{23}^B	Participant B divide equally with Participant C only *or* Participant B divide equally among all three participants
	E_{23}^C	Participant C divide equally with Participant B only *or* Participant C divide equally among all three participants
	E_{31}^B	Participant B divide equally among all three participants *or* Participant B divide equally with Participant A only
	E_{31}^C	Participant C divide equally among all three participants *or* Participant C divide equally with Participant A only
B	E_1^A	Participant A divide equally with Participant B only
	E_1^C	Participant C divide equally with Participant B only
	E_2^A	Participant A divide equally with Participant C only
	E_2^C	Participant C divide equally with Participant A only
	E_3^A	Participant A divide equally among all three participants
	E_3^C	Participant C divide equally among all three participants
	E_{12}^A	Participant A divide equally with Participant B only *or* Participant A divide equally with Participant C only
	E_{12}^C	Participant C divide equally with Participant B only *or* Participant C divide equally with Participant A only
	E_{23}^A	Participant A divide equally with Participant C only *or* Participant A divide equally among all three participants
	E_{23}^C	Participant C divide equally with Participant A only *or* Participant C divide equally among all three participants
	E_{31}^A	Participant A divide equally among all three participants *or* Participant A divide equally with Participant B only
	E_{31}^C	Participant C divide equally among all three participants *or* Participant C divide equally with Participant B only
C	E_1^A	Participant A divide equally with Participant C only
	E_1^B	Participant B divide equally with Participant C only
	E_2^A	Participant A divide equally with Participant B only
	E_2^B	Participant B divide equally with Participant A only
	E_3^A	Participant A divide equally among all three participants
	E_3^B	Participant B divide equally among all three participants
	E_{12}^A	Participant A divide equally with Participant C only *or* Participant A divide equally with Participant B only
	E_{12}^B	Participant B divide equally with Participant C only *or* Participant B divide equally with Participant A only
	E_{23}^A	Participant A divide equally with Participant B only *or* Participant A divide equally among all three participants
	E_{23}^B	Participant B divide equally with Participant A only *or* Participant B divide equally among all three participants
	E_{31}^A	Participant A divide equally among all three participants *or* Participant A divide equally with Participant C only
	E_{31}^B	Participant B divide equally among all three participants *or* Participant B divide equally with Participant C only

Table 1: List of events

Table 2: Normal form game under Total Deadweight Loss.

CA

	BA	BC	BAC
AB	(6,6,0)	(0,0,0)	(0,0,0)
AC	(6,0,6)	(6,0,6)	(6,0,6)
ABC	(0,0,0)	(0,0,0)	(4,4,4)

CB

	BA	BC	BAC
AB	(6,6,0)	(0,6,6)	(0,0,0)
AC	(0,0,0)	(0,6,6)	(0,0,0)
ABC	(0,0,0)	(0,6,6)	(4,4,4)

CAB

	BA	BC	BAC
AB	(6,6,0)	(0,0,0)	(4,4,4)
AC	(0,0,0)	(0,0,0)	(4,4,4)
ABC	(4,4,4)	(4,4,4)	(4,4,4)

Table 3: Normal form game under No Deadweight Loss.

CA

	BA	BC	BAC
AB	(6,6,0)	(4,4,4)	(6,4,2)
AC	(6,0,6)	(6,0,6)	(6,0,6)
ABC	(8,2,2)	(4,2,6)	(4,4,4)

CB

	BA	BC	BAC
AB	(6,6,0)	(0,6,6)	(2,8,2)
AC	(4,4,4)	(0,6,6)	(2,4,6)
ABC	(4,6,2)	(0,6,6)	(4,4,4)

CAB

	BA	BC	BAC
AB	(6,6,0)	(2,6,4)	(4,4,4)
AC	(6,2,4)	(2,2,8)	(4,4,4)
ABC	(4,4,4)	(4,4,4)	(4,4,4)